

POPLAR BLUFF, MO, USA

WMO: 723300

Lat: 36.773N

Lon: 90.325W

Elev: 100

StdP: 100.13

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03975

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.9	-7.8	-17.8	0.8	-8.8	-15.0	1.0	-6.2	11.0	8.6	9.7	8.6	2.9	320	0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	34.5	25.3	33.3	25.0	32.4	24.6	27.1	32.2	26.3	31.5	25.7	30.5	3.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.8	21.4	30.1	24.9	20.2	29.3	24.1	19.2	28.4	85.7	32.4	82.3	31.5	79.2	30.7	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	8.0	7.1	DB	-14.3	36.6	2.8	1.5	-16.3	37.7	-17.9	38.6	-19.5	39.5	-21.5	40.6	(4)
(5)			WB	-15.3	28.2	2.7	0.9	-17.2	28.8	-18.8	29.3	-20.3	29.8	-22.3	30.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.0	2.3	4.5	9.3	15.4	20.3	24.9	26.3	25.5	21.7	15.5	9.0	4.0	(6)	
	DBStd	9.64	5.85	5.95	5.53	4.51	4.04	2.81	2.50	2.47	3.56	4.62	4.88	5.34	(7)	
	HDD10.0	789	249	171	81	9	1	0	0	0	0	7	75	197	(8)	
	HDD18.3	2153	498	388	282	107	29	1	0	0	12	112	280	444	(9)	
	CDD10.0	2598	9	17	59	172	319	447	507	482	350	179	46	12	(10)	
	CDD18.3	919	0	0	3	21	89	198	248	223	112	25	1	0	(11)	
	CDH23.3	8854	0	1	21	168	750	2007	2531	2099	1007	258	13	0	(12)	
	CDH26.7	3559	0	0	1	26	219	852	1113	905	378	64	1	0	(13)	
(14) Wind	WSAvg	2.8	3.3	3.5	3.6	3.6	3.0	2.5	2.1	1.9	1.9	2.4	2.8	3.0	(14)	
(15) Precipitation	PrecAvg	1215	87	80	111	125	127	93	104	84	85	89	119	110	(15)	
	PrecMax	1650	158	202	294	422	268	185	213	295	171	342	317	246	(16)	
	PrecMin	674	27	27	54	0	0	25	24	12	12	1	30	18	(17)	
	PrecStd	219	40	38	54	79	56	42	52	57	44	68	73	60	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.0	21.9	25.9	29.0	32.7	35.8	36.3	36.1	34.1	31.1	25.4	19.8	(19)	
		MCWB	16.3	17.2	18.0	19.7	22.8	24.9	26.3	26.1	24.0	21.9	17.9	16.9	(20)	
	2%	DB	17.2	18.8	23.1	27.3	30.9	33.9	34.3	34.0	32.5	28.4	22.0	17.3	(21)	
		MCWB	15.0	14.6	17.3	19.2	22.4	24.5	26.0	25.3	23.7	20.2	16.0	14.7	(22)	
	5%	DB	14.2	16.8	20.7	25.3	29.0	32.6	33.1	32.8	30.9	26.4	19.6	14.9	(23)	
		MCWB	11.5	12.9	15.2	17.8	21.4	24.0	25.7	25.1	23.0	18.7	15.5	12.4	(24)	
	10%	DB	11.5	14.1	18.3	23.3	27.5	31.3	32.1	31.4	28.8	23.9	17.6	12.4	(25)	
		MCWB	8.9	10.8	13.7	17.1	20.8	23.5	25.5	24.5	21.8	17.6	13.8	10.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.3	18.1	19.6	21.9	24.7	26.9	28.0	28.0	25.8	23.5	20.0	17.8	(27)	
		MCDB	18.4	20.9	24.1	27.0	29.7	31.9	33.1	33.3	31.3	28.6	23.1	19.3	(28)	
	2%	WB	15.3	15.8	18.0	20.5	23.6	26.0	27.2	26.9	24.8	22.0	17.9	15.4	(29)	
		MCDB	17.3	18.1	22.1	25.1	28.4	31.6	32.3	32.0	30.2	26.3	20.3	17.2	(30)	
	5%	WB	12.4	13.9	16.4	19.4	22.7	25.3	26.6	26.1	23.9	20.5	16.3	12.8	(31)	
		MCDB	13.6	15.9	19.8	23.6	27.4	30.6	31.7	31.0	28.8	24.1	19.0	14.4	(32)	
	10%	WB	8.8	10.8	14.6	18.3	21.8	24.5	25.9	25.3	23.0	19.0	14.2	10.4	(33)	
		MCDB	11.1	13.7	17.5	22.0	26.3	29.5	30.7	30.0	27.2	22.7	17.1	12.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	9.9	10.8	11.6	10.9	10.8	10.0	10.7	12.4	12.9	11.2	8.8	(35)	
		MCDBR	11.9	13.2	13.6	13.3	12.4	12.1	11.0	12.3	14.0	15.2	14.2	11.2	(36)	
	5% WB	MCWBR	9.7	9.8	8.6	7.0	5.4	4.7	4.4	5.3	6.3	7.7	9.4	9.2	(37)	
		MCWBR	10.2	11.6	11.9	11.0	10.6	10.8	10.2	10.8	12.0	12.5	11.7	9.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.312	0.322	0.354	0.396	0.433	0.452	0.485	0.477	0.412	0.352	0.321	0.307	(40)	
		taud	2.485	2.446	2.403	2.282	2.239	2.195	2.142	2.153	2.321	2.474	2.511	2.508	(41)	
	Ebn at Noon		881	914	912	885	855	834	804	802	840	868	864	864	(42)	
		Edh at Noon	82	96	110	131	139	145	152	147	118	92	79	75	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.29	2.89	3.96	5.09	5.70	6.50	6.26	5.70	4.82	3.64	2.52	1.94	(44)	
	RadStd		0.22	0.37	0.33	0.34	0.47	0.42	0.33	0.32	0.29	0.40	0.27	0.20	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page